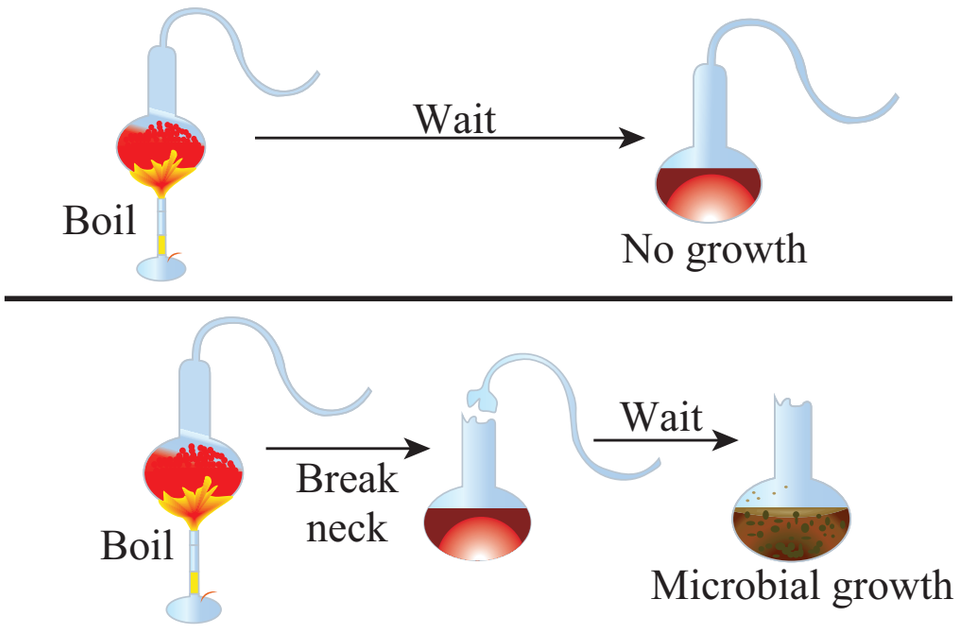


Figure 1.4: Pasteur's swan neck flask experiment

disease free silk worms. Wool Sorter's disease was named as "Anthrax" by him. He isolated *Bacillus anthracis* from the blood of infected animals. Chicken cholera bacterium was also isolated by Louis Pasteur using pure culture.

He proved that many diseases were caused by the presence of foreign microorganisms (**Germ theory of disease**). He discovered various infection causing microorganisms such as *Staphylococcus, Streptococcus* and *Pneumococcus.*

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Vaccination

Pasteur found out that bacteria could be attenuated by growing them in unnatural conditions. He coined the term "**attenuation**". It is a process wherein bacteria lose their virulence due to repeated subculturing under laboratory conditions. He used attenuated cultures as vaccines for immunizing and protecting an individual against the disease. He developed vaccines for anthrax and rabies.

1.2.3 Edward Jenner (1749-1823)

In ancient observation, persons who had suffered from a specific disease such as small pox (causative agent of small pox is varicella virus) or mumps, resisted the infection on subsequent exposures. They rarely contracted these infections for second time. Edward Jenner, a country doctor in England noted a pustular disease on the hooves of horses called the grease. This was carried by farm workers to the nipples of cows (cow pox). This was again carried by milk maids. They got inflamed spots on the hands and wrists. The people who got this cow pox were protected from small pox. He reported that 16 farm workers who had recovered from cow pox (causative agent of cow pox is vaccinia virus) were resistant to small pox infection.

He took the material (pus) from the cow pox and inoculated into the cut of 8 year old boy on 14th May 1796 (Figure 1.5). Two

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